

Experiment 9: Travelling Salesman Problem Aim:

Find the minimum tour cost.

Algorithm:

1. Start.
2. Apply the required AI technique.
3. Generate the solution.
4. Display the result.
5. Stop.

Python Code:

```
from itertools import permutations
```

```
graph = [
```

```
[0,10,15,20],
```

```
[10,0,35,25],
```

```
[15,35,0,30],
```

```
[20,25,30,0]
```

```
]
```

```
cities = [1,2,3]
```

```
min_cost = 999
```

```
for p in permutations(cities):
```

```
    cost = 0
```

```
    k = 0
```

```
    for j in p:
```

```
cost +=
```

```
graph[k][j]
```

```
k = j
```

```
cost += graph[k][0]
```

```
min_cost = min(min_cost,cost)
```

```
print("Minimum Cost =",min_cost)
```

Output

```
-----  
Best Path:  
(0, 1, 3, 2, 0)  
Minimum Cost: 80  
|
```

Result:

Minimum cost displayed.